Ziphozinle Mqopi

39/40

98.0/10

WIRE WAYS AND WIRING

SUMMATIVE ASSESMENTS

Module #	671101000-KM-05:
NQF Level	level 4,
Notional hours	110
Credit(s)	Cr 11
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

Details	Please Complete this Section
Name	Ziphozinle I
Surname	Mqopi
Cell NUMBER	084 045 6528
E-mail	mqopiziziphO@gmail.com

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Training	Classroom training registers	Training Provider Date:	
Self assessment	Assessment contract signed & dated	Learner file Date:	
Assessment contract	Assessment contract signed & dated	Learner file Date:	
Initial meeting	Assessor briefing checklist	Learner file Date:	
	Assessment contract & assessment plan	Learner file Date:	
Formative Assessment	Assessment instruments	Learner file Date:	
Summative Assessment	Assessment instruments	Learner file Date:	
Other Evidence	Research portfolio (if applicable)	Learner file Date:	
Feedback	Feedback Report	Learner file Date:	
Moderation	Moderators report	Learner file Date:	
Judgement	Assessor Summary Report / Moderator report	Learner file Date:	
1st Reassessment	Assessors summary report / instruments	Learner file Date:	
2nd Reassessment	Assessors summary report / instruments	Learner file Date:	

Special arrangements for assessment

Place
Language
Resources
Barriers

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- If not all evidence available was taken into account during the assessment
- The assessor was not a subject matter expert or did not have a subject matter expert during the assessment process
- The assessor did not assess according to the performance criteria and range statement stipulated in the unit standard
- Not all the range items were available for assessment

2. A learner bringing an appeal against a decision of the assessment will lodge such an appeal with the assessor and the internal moderator within 2 days of the assessment feedback session.

3. A learner bringing an appeal should complete the "Learner's Notice of Assessment Appeal "form before the Appeal Hearing. The form should be handed to the internal moderator or a representative of the SETA.

4. Should the internal moderator re-affirm the assessor' decision, the learner may appeal to the external verifier within 2 days after the initial moderator's feedback session. The external verifier's decision will be final. Should the external verifier re-affirm the assessors' decision, the cost for re-evaluation will be upon the learner. Should the verifier's decision differ from the assessor's decision, the cost for re-evaluation will be borne by the assessor.

EASTC TECHNOCENTRIC VARSITY

ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

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Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____ Date: _____

Employee witness: _____ Date: _____

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Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: Ziphozinhle Maopi	Signature
Assessors Name: Balenah Yabadi	Signature:

INSTRUCTIONS:

The following sections will test your knowledge and you are required to answer all the questions.

All true and false items must be answered.

Once you have finished you may obtain feedback from your assessor on the accuracy of your answer.

2) Drip- Tight Wireway (NEMA 2)
Designed with overlapping seams or drip shields to prevent water droplets from entering the enclosure when mouted in a vertical or near-vertical position. Suitable for indoor environments where dripping water or condensation may be present. Examples include areas with overhead piping

3.
1) Conduit : Protects wires in exposed areas, outdoors, or industrial setting, available in Rigid Metal Conduit for heavy-duty, Intermediate Metal Conduit for medium-duty, PVC conduit for corrosion resistance, and flexible conduit for bends and twists
2) Tranking : Managing multiple cables in offices, data centres centers, or industrial spaces with Open Ladder for flexibility, Closed Solid Bottom for protection and Ventilated for airflow and cable management
3) Ducting : Conceals wiring under floors, in walls, or surface-mouted with Split Ducting for easy acess, Solid Ducting for secure installations and underfloor Ducts for raised floor applications
4) Gouiding : Enables surface wiring in homes or offices without damage, available in plastic for low voltage wiring. Metal for durability and heavy cables and Mini Trans for discreet, aesthetic profiles
5) Wire Mould : Used in commercial spaces for minimal disruption with Metal steel or Aluminium for durability and industrial look and Plastic for flexibility and various colour options

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Task	Questions Description	Marks
3	a) Describe the principles of operation of various control systems b) Describe the principles of operation of single and three phase circuit breakers and core balance earth leakage relays (wound primaries and straight primaries with tripping relay)	5 5

3a)

Input: The system receives an input signal, which can be a command, a measurement, or a combination of both. This input is crucial as it determines the system's behaviour

Process: The system processes the input signal to produce an output signal. This process can involve calculations, transformations or adjustments based on the input.

Output: The system produces an output signal, which can be physical action a change in state, or signal to another system.

Feedback: In some systems, feedback is used to compare the actual output to the desired output. This allows the system to make adjustments to maintain the desired output

Control: The system is designed to control the output based on the input and the process This involves setting up mechanisms to ensure that the output remains within a desired range.

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b.) Single-Phase Circuit Breakers: Interrupt fault currents in single-phase circuits using thermal or magnetic trips Bimetal strips bend with heat from overloads tripping the breaker. High fault currents trigger an electromagnet to trip the breaker.

Three-Phase Circuit Breakers: Iterrupt faults in three-phase systems often using common trip mechanisms. Typically trip all phases if one phase detects a fault ~~et~~ ensuring isolation

Core Balance Earth Leakage (Wound Primaries) Relays: Measure ~~resind~~ residual current via a current transformer around conductors. If an imbalance exceeds threshold, the relays trips, isolating the circuit.

Core Balance Earth Leakage Relays (Straight Primaries): Similar detection principles but often directly trip breakers via a sensitive relay. It has fast response to earth faults protecting against shocks and fires.

2

c)

1) Underground cables:

Armouring / Sheath Earthing: Metallic armour / sheath earthed at both ends to provide a low-resistance path for fault currents, reducing risk of damage or shock.

Earth continuity: Ensures continuity of earth path even if cables are damaged

It minimizes risk of electrical shock or cable damage due to faults

2) Overhead cables

Earth Wires

Positioned above phase conductors, protect against lightning strikes and induced surges

Structure Earthing: Metal poles, towers, and fittings earthed to ensure safety

It minimizes step/touch pontentials reducing risk of electric shock during faults.

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Assessor's Report

Candidate's Name	Ziphozihle Maqpi	ID No.	0802030118086
Assessor's Name	Baienah Tjabadi	Reg. No.	namb-A/2025/5307
Title	Wire ways and Wiring		

ASSESSMENT DECISION

Outcome	C	NYC	Comments

OVERALL COMPETENCE JUDGEMENT:

Comments

~~Good work~~

Date	
Signature of Assessor	Signature of Candidate

Candidate Appeal Form

Candidate's Name:		ID No.	
Assessor's Name:		Reg. No.	
Title:			
Date:			
SECTION 1			
Candidate's reason for disagreeing with the assessment decision			
Assessor's rationale for the assessment decision			
Candidate's signature			
Assessor's signature			
SECTION 2			
Moderator's decision			
Moderator's signature			

EASTC TECHNOCENTRIC VARSITY

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ASSESSMENT REVIEW

NAME of LEARNER Ziphozihle Maopi	NAME of ASSESSOR Balennah Jabadi
VENUE EASTC TECNOCENTRIC VARSITY	DATE of REVIEW

Review Dimension	ASSESSOR	LEARNER/ CANDIDATE	ACTION
The principles/criteria for good assessment were achieved?	☑ Agree ☐ Disagree	☑ Agree ☐ Disagree	
The assessment related to the registered unit standard?	☑ Agree ☐ Disagree	☑ Agree ☐ Disagree	
The assessment was practical?	☑ Agree ☐ Disagree	☑ Agree ☐ Disagree	
It was time efficient and cost-effective and did not interfere with my normal responsibilities?	☑ Agree ☐ Disagree	☑ Agree ☐ Disagree	
The assessment instruments were fair, clear and understandable	☑ Agree ☐ Disagree	☑ Agree ☐ Disagree	
The assessment judgements was made against set requirements	☑ Agree ☐ Disagree	☑ Agree ☐ Disagree	
The venue and equipment was functional?	☑ Agree ☐ Disagree	☑ Agree ☐ Disagree	
Special needs were identified and the assessment plan was adjusted	☑ Agree ☐ Disagree	☑ Agree ☐ Disagree	
Feedback was constructive against the evidence required	☑ Agree ☐ Disagree	☑ Agree ☐ Disagree	
An opportunity to appeal was given	☑ Agree ☐ Disagree	☑ Agree ☐ Disagree	
The evidence was recorded	☑ Agree ☐ Disagree	☑ Agree ☐ Disagree	

LEARNER'S DECLARATION OF UNDERSTANDING

I am aware of the moderation process and understand that the moderator could declare the assessment decision invalid

Learner Date	Assessor Date	Moderator Date

Z